Descriptive Analytics Report: Fashion Trends at Scale

Tracking Trending Styles, Sales Velocity, Ratings, Regional Demand & Inventory

Note on data: This report uses the supplied Fashion Trend Dataset workbook. The workbook contains 300 category-day observations across 10 categories, 8 regions, 10 colors, and 60 tracked SKUs. The 30-day sales data records 2,054,556 units and $151,065,686.26 in revenue.

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1. Objective

Build a descriptive analytics layer over daily fashion purchase transactions to answer the recurring business questions that a merchandising, buying, or trend-forecasting team asks every day:

1. What styles and products are trending or best-selling right now?
2. How are total sales trending day over day, and what weekly or seasonal rhythm sits underneath that trend?
3. Which categories, silhouettes, or style attributes are gaining or losing popularity over the course of the month — i.e., what is rising and what is fading?
4. How are products rated by customers, and where is rating quality slipping in a way that signals a fit, fabric, or fulfillment problem?
5. Where is fashion demand concentrated geographically, and does that concentration match where inventory actually sits?
6. Which SKUs are at risk of stocking out, and are those the same SKUs currently driving the trend?
7. How do price point and discount depth relate to sell-through — are markdowns actually accelerating trend adoption, or just eroding margin?
8. Which colors, prints, and materials are trending upward or downward, independent of any single product — the signal a design team needs before the next buy cycle?

2. How This Would Work at Real Scale

Processing daily purchases across a large fashion catalog is a data-pipeline problem before it's an analytics problem — but at the scale most fashion retailers operate at, it does not require streaming infrastructure. A conventional, batch-oriented setup is normal and sufficient:

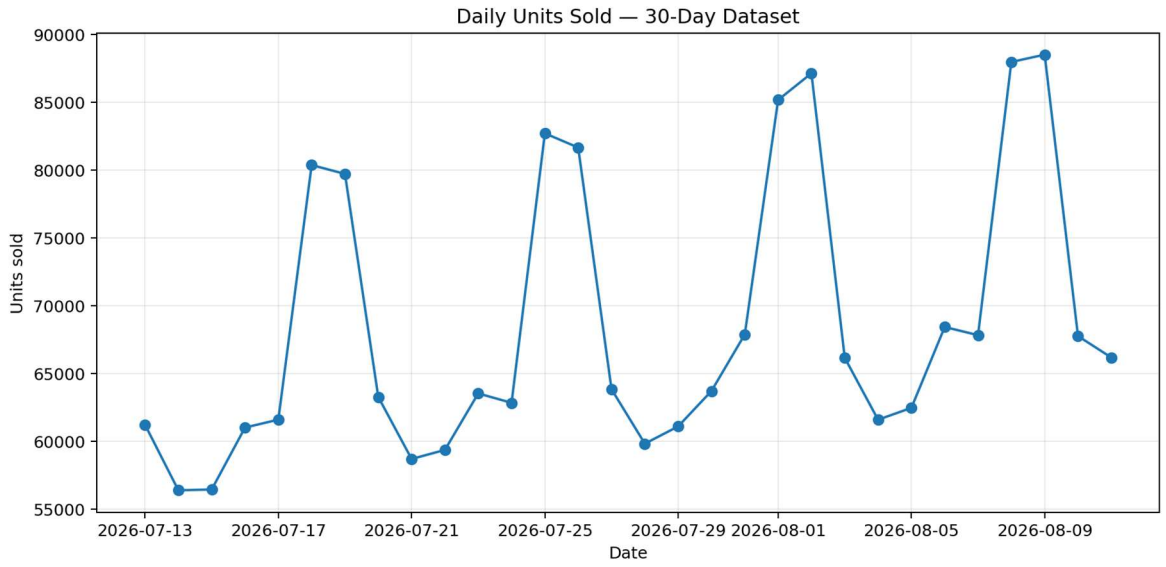
Layer	Purpose	Typical tools
Data capture	Record each purchase, return, and inventory update as it happens at checkout or in-store POS	Point-of-sale system / e-commerce platform export (Shopify, Magento, or in-house checkout database)
Storage	Hold raw transaction records and product/style metadata in one queryable place	Relational database (MySQL / PostgreSQL) or a shared data warehouse (Snowflake / BigQuery)
Cleaning & prep	Standardize product attributes (category, color, material), handle returns/cancellations, de-duplicate records	Python (pandas) or SQL scripts run on a schedule
Aggregation	Roll raw rows up into daily, per-category, per-region, and per-style summary tables	SQL queries / Excel pivot tables for smaller catalogs, dbt or scheduled scripts for larger ones
Scheduling	Run the cleaning and aggregation steps automatically once a day	Cron job or Windows Task Scheduler, or a lightweight workflow tool if the team already has one
Reporting / BI layer	Turn the aggregated tables into dashboards the merchandising and buying teams check daily	Power BI, Tableau, Google Data Studio, or well-built Excel dashboards

The report below reflects the output of that pipeline — daily aggregated tables — not raw transaction-level data, since no dashboard renders millions of individual line items usefully. Everything downstream (the charts and tables described in the sections that follow) is built on top of these daily summaries.

3. Daily Sales Trend

The Daily Sales Trend graph shows total units sold for each of the 30 dates in the dataset. The dataset records **2,054,556 units** and **$151,065,686.26 revenue** across the reporting window. The graph shows noticeable variation in daily sales throughout the 30-day period. The highest daily sales were 88,501 units, while the lowest daily sales were 56,402 units. The 32,099-unit difference indicates variation in daily demand. The trend helps identify high-demand and low-demand periods for further sales and inventory analysis.

Graph



KPI Definitions Used

- Total Units Sold: The total number of units sold during the 30-day reporting period

- Total Revenue: The total revenue generated from sales during the reporting period.

- Highest Daily Sales: The maximum number of units sold on a single day during the reporting period.

- Lowest Daily Sales: The minimum number of units sold on a single day during the reporting period.

- Peak-to-Low Difference: The difference between the highest and lowest daily units sold, indicating the variation in daily sales volume

- Daily Sales Trend: The day-by-day movement of units sold over the 30-day period, helping identify high-demand and low-demand dates.

Key numerical summary

Measure	Value
Reporting period	30 days
Total units sold	2,054,556 units
Total revenue	$151,065,686.26
Highest daily units	88,501 units (09 Aug 2026)
Lowest daily units	56,402 units (14 Jul 2026)
Peak-to-low difference	32,099 units

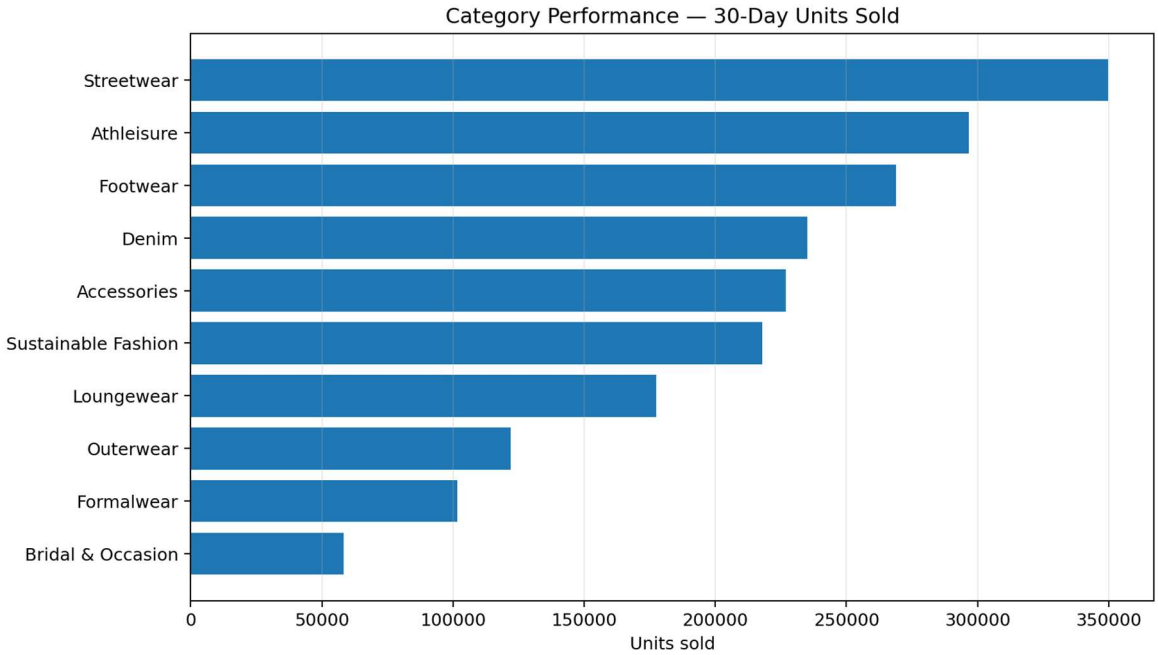
Detailed interpretation

The highest daily sales volume is **88,501 units** on **09 Aug 2026**, while the lowest is **56,402 units** on **14 Jul 2026**. The difference of **32,099 units** demonstrates meaningful day-to-day variation. Higher points represent stronger observed demand and lower points represent weaker observed demand. This graph helps identify peak dates, slower periods, and unusual movements that may need investigation. It does not by itself prove that a particular event, discount, or marketing activity caused a spike.

4. Category Performance

This section replaces the earlier 'Trending & Best-Selling Styles (Most Recent Day)' template because the supplied workbook supports **category-level 30-day performance**, not a top-15/20 SKU leaderboard for one day. The graph compares total units sold across the 10 categories. The Category Performance graph compares sales volume across the 10 fashion categories. Streetwear recorded the highest volume with 349,823 units, while Bridal & Occasion recorded the lowest with 58,350 units. The difference of 291,473 units shows a significant variation in category-level demand.

Graph



Key numerical summary

Category	30-day units	30-day revenue
Streetwear	349,823	$14,678,479.44
Bridal & Occasion	58,350	$18,267,127.65
All categories	2,054,556	$151,065,686.26

KPI DEFINE USED:

- *Measures total units sold across different product categories.*
- *Helps compare the performance of different fashion categories.*
- *Identifies the best-performing and weakest categories.*
- *Shows which categories contribute most to overall sales.*
- *Supports product planning and category-level decision-making.*

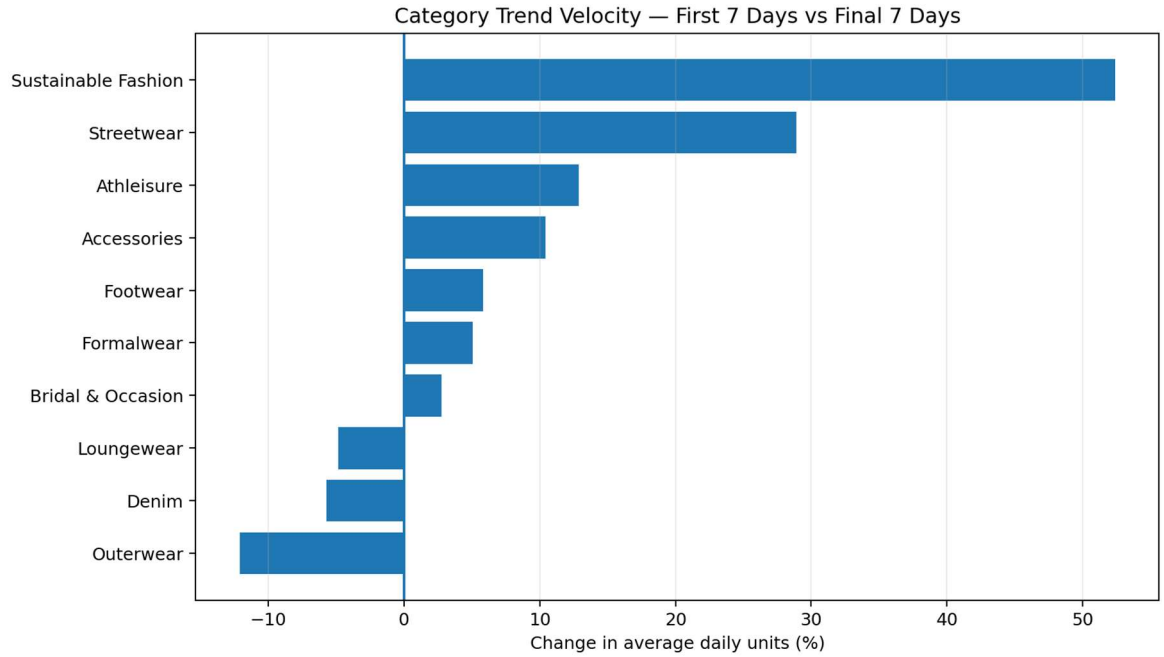
Detailed interpretation

Streetwear is the highest-volume category with **349,823 units**, while **Bridal & Occasion** has the lowest with **58,350 units**. The difference is **291,473 units**. The bars show where demand is concentrated across categories. A larger bar means that category contributed more units during the 30-day period. This supports category comparison and descriptive merchandising or inventory discussions, but it does not explain why one category performs better.

5. Style & Category Trend Velocity

Trend velocity measures change over time rather than simply ranking categories by total volume. For this dataset, it compares average daily units during the first seven days with the final seven days. Sustainable Fashion showed the strongest positive movement, increasing by 52.4%, from an average of 5,959.6 units/day in the first seven days to 9,082.4 units/day in the final seven days. Outerwear showed a decline of 12.1%, from 4,329.4 units/day to 3,805.1 units/day. This indicates a stronger recent demand for Sustainable Fashion and a decline in demand for Outerwear during the observed period

Graph



Key numerical summary

Category	First 7-day avg.	Last 7-day avg.	Change
Sustainable Fashion	5959.6	9082.4	+52.4%
Outerwear	4329.4	3805.1	-12.1%

KPI DEFINE USED:

- *Measures the rate of change in sales for different categories.*
- *Identifies categories experiencing increasing or decreasing demand.*
- *Helps detect fast-growing fashion categories.*
- *Shows changes in customer preferences over time.*
- *Supports timely decisions related to product selection and inventory.*

Detailed interpretation

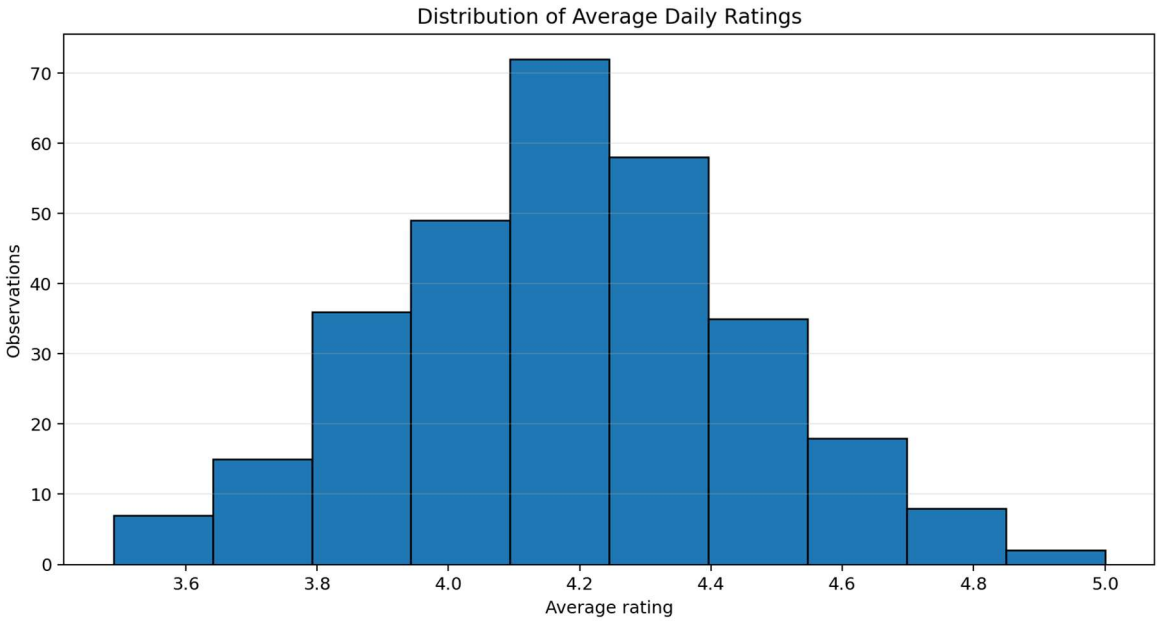
Sustainable Fashion has the strongest positive movement at **+52.4%**, increasing from **5959.6 units/day** to **9082.4 units/day**. **Outerwear** has the largest decline at **-12.1%**, moving from **4329.4 units/day** to **3805.1 units/day**. Positive change indicates stronger recent average demand, while negative change indicates weakening recent average demand. These are descriptive changes within the observed period, not forecasts.

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6. Rating & Review Distribution

The Rating Distribution graph shows how reported average daily ratings are distributed. The overall mean rating is **4.19/5**.The average rating is 4.19/5, with observed ratings ranging from 3.49 to 5.00 across 300 rating observations. The distribution helps understand overall customer rating patterns and whether ratings are concentrated around higher or lower values. □

Graph



Key numerical summary

Rating metric	Value
Average rating	4.19 / 5
Minimum observed rating	3.49
Maximum observed rating	5.00
Rating observations	300

KPI DEFINE USED:

- *Measures the distribution and average level of customer ratings.*
- *Helps evaluate overall customer satisfaction with products.*
- *Identifies products or ratings receiving stronger customer responses.*
- *Shows the general perception of product quality.*
- *Supports improvement in product quality and customer experience.*

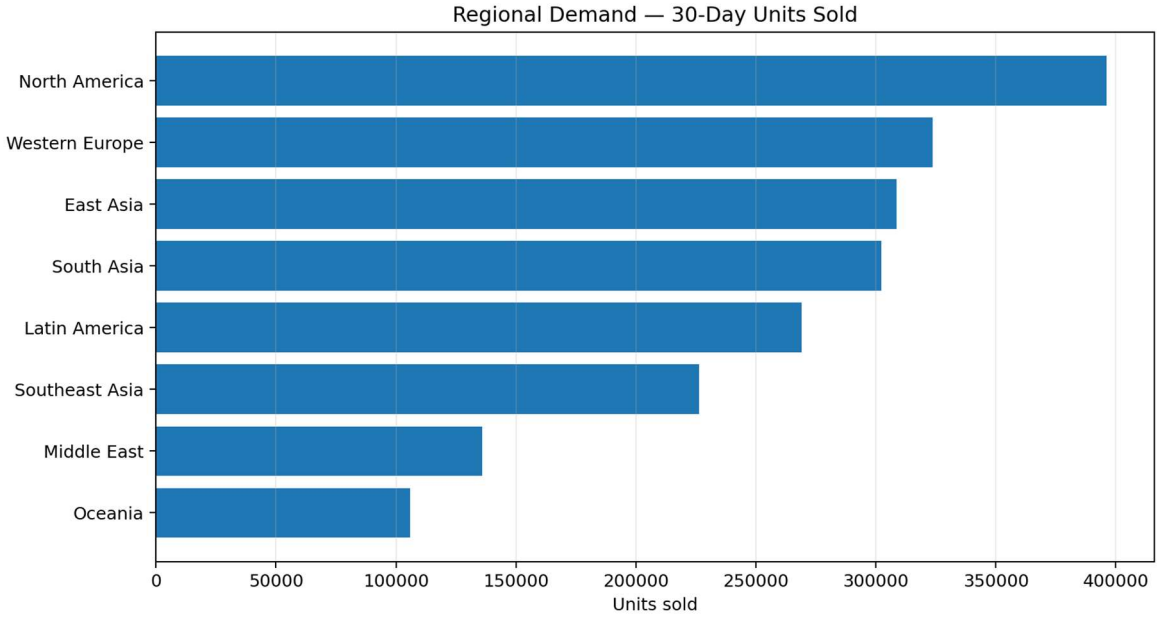
Detailed interpretation

The average rating of **4.19/5** provides an overall summary, while the histogram shows the distribution behind that average. The observed ratings range from **3.49** to **5.00**. More observations in a rating range produce a taller bar. This helps show whether ratings are concentrated or widely spread. The graph describes rating patterns and does not establish that ratings cause higher or lower sales.

7. Regional Demand

The Regional Demand graph compares 30-day unit sales across the **8 regions** in the workbook. The Regional Demand graph compares unit sales across the 8 regions in the dataset. North America recorded the highest demand with 396,122 units (19.16%), while Oceania recorded the lowest with 105,794 units (5.12%). The difference of 290,328 units shows that demand is not evenly distributed across regions.

Graph



Key numerical summary

Region	30-day units	Share of total
North America	396,122	19.16%
Oceania	105,794	5.12%
All regions	2,067,909	100.00%

KPI DEFINE USED:

- *Measures the number of products sold across different regions.*
- *Helps identify regions with high and low product demand.*
- *Shows geographical differences in customer purchasing behaviour.*
- *Supports region-wise sales and marketing planning.*
- *Helps businesses focus inventory on areas with stronger demand.*

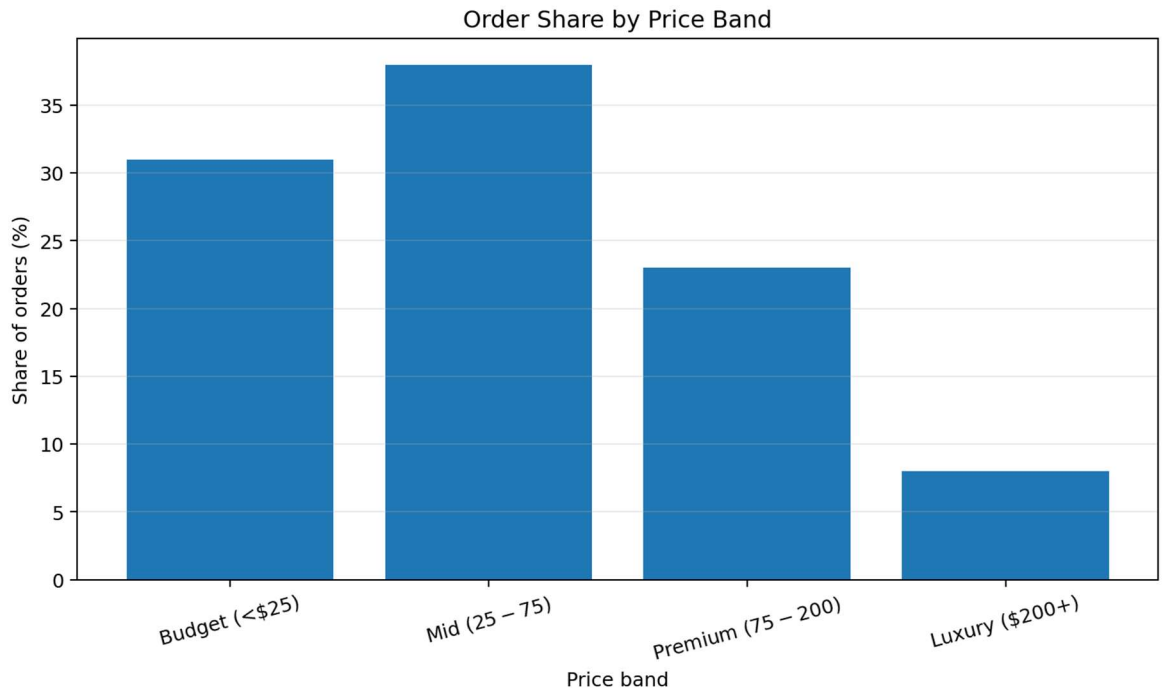
Detailed interpretation

North America records the highest regional demand with **396,122 units**, equal to **19.16%** of the total. **Oceania** has the lowest with **105,794 units**, or **5.12%**. The difference is **290,328 units**. The graph shows that demand is not evenly distributed across regions and can support descriptive regional planning.

8. Price Point & Order Share

The workbook contains order share by price band, but it does not contain SKU-level discount-event data. Therefore, a discount-correlation chart or a claim that a particular discount caused higher sales should not be included. The Mid ($25–$75) price band has the highest order share at 38.0%, followed by the Budget (<$25) band at 31.0%. The Luxury ($200+) category has the lowest order share at 8.0%. This indicates that the mid-price range represents the largest share of orders in the dataset.

Graph



Key numerical summary

Price band	Order share
Budget (<$25)	31.0%
Mid ($25-$75)	38.0%
Premium ($75-$200)	23.0%
Luxury ($200+)	8.0%

KPI DEFINE USED :

- *Measures the percentage of orders within different price ranges.*
- *Identifies the price bands preferred by customers.*
- *Helps understand customers' spending behaviour.*
- *Shows which price range contributes the highest order share.*
- *Supports pricing and product positioning decisions.*

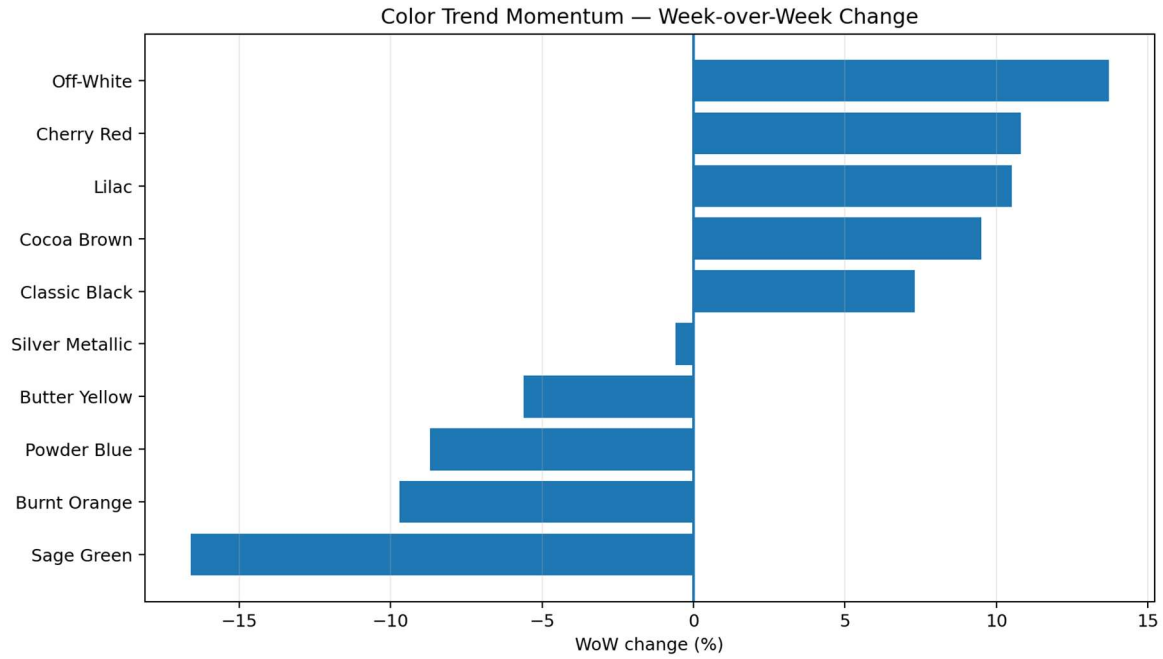
Detailed interpretation

The largest share of orders is in **Mid ($25-$75)**, representing **38.0%** of orders. The smallest share is **Luxury ($200+)**, representing **8.0%**. The difference is **30.0 percentage points**. This is an order-share by price-band analysis, not a discount-correlation analysis, because the workbook does not provide discount-event observations.

9. Color, Print & Material Trends

The supplied Color Trends sheet contains order share and week-over-week momentum for ten colors. The graph therefore focuses on the color-level trend movement directly supported by the workbook. The color analysis shows that Off-White recorded the strongest week-over-week increase of 13.7%, while Sage Green recorded the largest decline of 16.6%. The highest order-share color represented 14.5% of orders. This analysis helps identify colors gaining or losing demand during the observed period

Graph



Key numerical summary

Color	Order share	WoW change
Off-White	7.4%	+13.7%
Sage Green	8.3%	-16.6%
Highest order-share color	14.5%	9.5%

KPI DEFINE USED:

- *Measures sales performance and demand for different product colors.*
- *Identifies colors that are becoming more or less popular.*
- *Helps understand changing fashion and customer preferences.*
- *Supports decisions about future color selection.*
- *Helps businesses plan products according to current color trends.*

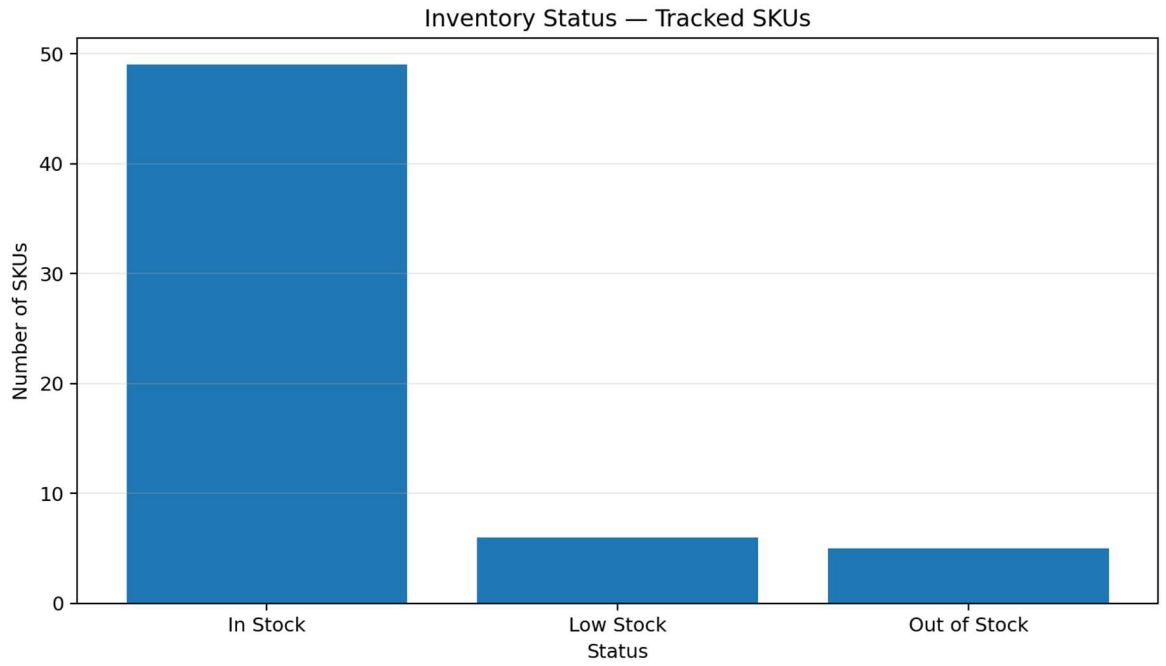
Detailed interpretation

Off-White has the strongest week-over-week increase at **+13.7%**, while **Sage Green** has the largest decline at **-16.6%**. The difference is **30.3 percentage points**. Order-share values also show the relative importance of each color. This is an attribute-level trend signal and does not prove that color alone caused sales changes.

10. Inventory Status

The Inventory Status graph summarizes the availability status of the **60 tracked SKUs**. The workbook provides SKU status and units left, so the supported chart is a status-count comparison rather than a days-of-supply calculation. The Inventory Status analysis shows that 49 SKUs (81.7%) are In Stock, while 6 SKUs (10.0%) are Low Stock and 5 SKUs (8.3%) are Out of Stock. Overall, 11 SKUs (18.3%) are classified as At-Risk Inventory and may require attention for inventory planning.

Graph



Key numerical summary

Inventory status	SKU count	Share of tracked SKUs
In Stock	49	81.7%
Low Stock	6	10.0%
Out of Stock	5	8.3%
Low + Out of Stock	11	18.3%

KPI DEFINE USED:

- *Measures the availability of products across different inventory statuses.*
- *Identifies products that are adequately stocked or require attention.*
- *Helps monitor potential stock shortages.*
- *Supports better inventory management and replenishment planning.*
- *Helps maintain product availability according to customer demand.*

Detailed interpretation

There are **49 In Stock SKUs**, **6 Low Stock SKUs**, and **5 Out of Stock SKUs**. Together, **11 SKUs**, or **18.3%** of tracked inventory, are either Low Stock or Out of Stock. This gives a clear numerical picture of stock availability. The graph becomes more useful when read alongside category and trend graphs. Because the workbook does not provide a per-SKU daily sell-through series, exact days of supply should not be claimed from this sheet alone.

11. Summary of Key Metrics

Metric	Value (simulated)
Total units sold (30 days)	2,054,556
Total revenue (30 days)	$151,065,686.26
Average product rating	4.19 / 5
Top single-day seller	88,501 units (09 Aug 2026)
Fastest-rising category	Sustainable Fashion (+52.4%)
Fastest-declining category	Outerwear (-12.1%)
Largest order-share price band	Mid ($25-$75) (38.0%)
Regions tracked	8
Low Stock + Out of Stock	11 of 60 SKUs

12. Next Steps to Move From Descriptive → Predictive

This report answers what happened and what is currently trending. The natural next layer, if useful:

- Demand forecasting per SKU/region to set reorder points automatically ahead of a trend peak rather than reacting after it.
- Trend-lifecycle modeling that flags whether a rising style is still early, near its peak, or already fading, so buying decisions are timed to the curve rather than to a single day's snapshot.
- Anomaly detection on rating drops or sudden demand spikes, so a quality issue or a viral moment surfaces automatically instead of waiting for a human to notice it in a dashboard.
- Cohort and repeat-purchase analysis, to separate genuinely new demand from existing customers simply buying more of a trending category.

Happy to build out any of these next, or swap in a real dataset (POS export, e-commerce database connection, CSV/Excel file) if you have one — the same tables and chart structure above would apply directly.

SUBMITTED BY

Name: UNNATI SINGH

Roll No.: 1250258487(39)

THANKYOU